JAMALIA SULTANA

Ph.D. Candidate, Computer Science

Stony Brook University | Advisor: Prof. Zhaozheng Yin

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Research Focus

Computer vision and multimodal learning from human expertise, using gaze, speech, and interactions to guide medical and egocentric image understanding. First-author work at **WACV, ISBI (oral), and ACCV**.

Education

Stony Brook University, USA — Aug. 2023 – Present
Ph.D. in Computer Science; Ph.D. Candidate

Bangladesh University of Engineering and Technology (BUET), Dhaka
M.Sc. Engg. in Computer Science and Engineering — Jul. 2021 – Jun. 2023
B.Sc. in Computer Science and Engineering — Feb. 2016 – Feb. 2021

Research Experience

Graduate Research Assistant — Jun. 2024 – Present
Stony Brook University | Supervisor: Prof. Zhaozheng Yin

- **EgoWaste (WACV 2027).** Built an egocentric waste dataset and a semi-automatic annotation pipeline combining expert speech, foundation-model proposals, tracking, and hand-object interaction, with human verification. Provides **35,754 frames and 39,083 active-object instances** across 15 material categories and 56 fine-grained classes. Project repository.
- **Learning from radiologists.** Developed gaze- and speech-based graph learning for chest X-ray classification (**ACCV 2024**) and anatomically grounded report generation (**ISBI 2026, oral**). Developed gaze-informed representation learning for downstream classification, localization, and segmentation (**TMI, under review**).
- **Gaze as action.** Developing an action-conditioned world model that predicts evidence from candidate fixations and updates a diagnostic belief state; investigating observation selection and stopping with fewer looks.
- **AI for science, BNL collaboration.** Developing sparse X-ray diffraction reconstruction with Bragg-spot evaluation, and geometric learning for protein binding-site localization.

Graduate Research Fellow — Jul. 2021 – Jun. 2023
BUET | Supervisors: Prof. Mahmuda Naznin and Prof. Tanvir Faisal

- Developed semi-supervised 3D femur segmentation from QCT images for patient-specific reconstruction (**MBEC 2024; ICMLA 2023**).

Graduate Research Assistant — Aug. 2021 – Jun. 2022
United International University, Bangladesh

- Studied image-based food recognition, volume estimation, and nutrient inference; synthesized datasets, methods, and evaluation gaps (**IEEE Access 2023**).

Mentoring & Teaching

Research Mentor, Stony Brook University — Summer 2026

- Mentored undergraduate researchers through **AI3 REU** and high school researchers through **CSIRE**, guiding computer vision projects, experiments, and research communication.
- Supported a CSIRE project, "Mining Egocentric Video with Sorter Speech for Plastic Classification," accepted to the **IEEE BigData 2026 High School Symposium**.

Teaching: Guest presenter, BMI 514: Imaging Informatics Analysis; teaching assistant, CSE 334: Introduction to Multimedia and CSE 310: Computer Networks.

Publications

1. **Jamalia Sultana**, Ruwen Qin, and Zhaozheng Yin. "EgoWaste: A Hierarchical Egocentric Waste Dataset for Material Recovery." IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2027.
2. **Jamalia Sultana**, Ruwen Qin, and Zhaozheng Yin. "RadGaze-LLM: Anatomical Region-Grounded Radiology Report Generation via Learning from Expert Gaze." IEEE International Symposium on Biomedical Imaging (**ISBI**), 2026. **Oral presentation.**
3. **Jamalia Sultana**, Ruwen Qin, and Zhaozheng Yin. "Seeing Through Expert's Eyes: Leveraging Radiologist Eye Gaze and Speech Report with Graph Neural Networks for Chest X-ray Image Classification." Asian Conference on Computer Vision (**ACCV**), 2024.
4. **Jamalia Sultana**, Mahmuda Naznin, and Tanvir Faisal. "SSDL–An efficient automated semi-supervised deep learning approach for patient-specific 3D reconstruction of proximal femur from QCT images." **Medical & Biological Engineering & Computing**, 2024.
5. **Jamalia Sultana**, Mashiyat Nayeem, Mahmuda Naznin, and Tanvir Faisal. "Semi-Supervised Learning Based Femur Segmentation from QCT Images." IEEE International Conference on Machine Learning and Applications (**ICMLA**), 2023.
6. **Jamalia Sultana**, Benzir Md. Ahmed, Mohammad Mehedy Masud, A. K. Obidul Huq, Mohammed Eunus Ali, and Mahmuda Naznin. "A Study on Food Value Estimation From Images: Taxonomies, Datasets, and Techniques." **IEEE Access**, 2023.
7. **Jamalia Sultana** and Mahmuda Naznin. "Breaking the Barrier with a Multi-Domain SER." IEEE Annual Computers, Software, and Applications Conference (**COMPSAC**), 2022.

Manuscript Under Review

Jamalia Sultana, Ruwen Qin, and Zhaozheng Yin. "Gaze-conditioned Pretraining for Representation Learning in Chest X-ray Images." **IEEE Transactions on Medical Imaging**. Submitted October 2026; under review.

Academic Service & Recognition

Peer reviewer: CVPR, ICCV, NeurIPS, ACCV, CVMI.

Champion Student Poster Award, NSysS 2022. Research on fault tolerance in deep learning under human-induced medical data corruption.

ACM Student Research Competition selection, Grace Hopper Celebration, 2020. Research on biological gender and emotion detection in voice data.

Selected Research Presentations

Stony Brook Cancer Center Annual Research Symposium, 2026. "Learning Clinical Expertise from Radiologists' Eye Gaze and Speech."

CMBBE 2023; Orthopaedic Research Society 2023. Presented research on automated femur segmentation and patient-specific 3D reconstruction.

Technical Expertise

Vision & learning: PyTorch, TensorFlow, SAM/SAM2, CLIP, Hugging Face, OpenCV.
Scientific computing: NumPy, pandas, Matplotlib, RDKit, OpenMM, MSMS.
Programming: Python, MATLAB, C/C++, Java, SQL, JavaScript.